GREEN CITY

Portfolio Management

Learn how to manage and adapt an evolving portfolio as markets, risks and information change.

START

Understand each new portfolio tool.

BUILD

Calculate before making decisions.

CAPABILITY

Adapt without abandoning discipline.

Where Green City fits

Green City assumes the portfolio already exists. Your job is now to manage the whole system.

PURPLE CITY

Learn how to invest: foundations.

BLUE CITY

Learn what to invest in: analysis and selection.

GREEN CITY

Learn how to manage and adapt the portfolio.

YELLOW CITY

Prove you can manage professionally.

How to learn from this deck

Every calculation is taught as a decision tool, not as isolated mathematics.

1 UNDERSTAND

What does the idea mean?

2 FORMULA

What relationship are we measuring?

3 SUBSTITUTE

Where do the actual numbers go?

4 CALCULATE

Show every important step.

5 INTERPRET

Translate the answer into plain English.

6 MANAGE

What does it change in the portfolio?

Green City roadmap

Nine portals turn individual investments into an actively managed portfolio.

19 Bonds	20 ETFs & Alternatives	21 Allocation	22 Diversification	23 Add / Trim / Exit	24 Macro	25 New Information	26 Risk	27 Crisis

The same portfolio continues. New knowledge changes how you understand, size, protect and revise it.

The Green City management mindset

A good asset can still create a bad portfolio if it is badly sized or badly combined.

RETURN

What can this holding contribute?

RISK

How can it hurt the portfolio?

LIQUIDITY

Can capital be accessed when needed?

CORRELATION

What else tends to move with it?

CASH FLOW

Does it generate or consume cash?

CONCENTRATION

Is too much dependent on one idea?

NEW INFORMATION

Has the evidence changed?

ACTION

Add, hold, trim, exit, replace, hedge or rebalance?

PORTAL 19

Bonds & Fixed Income

Understand government and corporate bonds, yields, duration, credit risk and interest-rate sensitivity.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Bond basics

A bondholder is mainly a creditor, not an owner.

BOND

A loan made by the investor to a government or company.

FACE VALUE

Principal normally repaid at maturity.

COUPON RATE

Contractual interest rate applied to face value.

MATURITY

Date principal is due.

MARKET PRICE

What investors pay for the bond today.

YIELD

A return measure linked to price and cash flows.

Coupon payment

Face value $1,000; annual coupon rate 6%.

FORMULA

$$\text{Annual coupon} = \text{Face value} \times \text{Coupon rate}$$

WHAT ARE WE MEASURING?

Contractual annual cash interest.

WORKINGS

6% = 0.06
Coupon = $1,000 × 0.06
Coupon = $60

ANSWER

$60 per year

PORTFOLIO MEANING

Useful for income planning, but market value can still rise or fall.

Current yield

Annual coupon $60; bond price $950.

FORMULA

$$\text{Current yield} = \text{Annual coupon} / \text{Market price} \times 100\%$$

WHAT ARE WE MEASURING?

Coupon income relative to today's purchase price.

WORKINGS

$60 / $950 = 0.0631579
0.0631579 × 100% = 6.31579%

ANSWER

≈ 6.32%

PORTFOLIO MEANING

Not the same as total return or yield to maturity because it ignores maturity value and timing.

Holding-period return

Buy at \$980; receive \$50 coupon; sell at \$1,005.

FORMULA

$$\text{HPR} = (\text{Coupon} + \text{Ending price} - \text{Beginning price}) / \text{Beginning price} \times 100\%$$

WHAT ARE WE MEASURING?

Coupon income plus price change over the holding period.

WORKINGS

Price gain = \$1,005 − \$980 = \$25
Total gain = \$50 + \$25 = \$75
\$75 / \$980 × 100% = 7.65306%

ANSWER

≈ 7.65%

PORTFOLIO MEANING

Bond return is not only the coupon. Market-price movement matters.

Why bond prices and yields move in opposite directions

This inverse relationship is one of the most important fixed-income ideas.

RATES RISE

New bonds can offer more attractive returns. Existing lower-paying bonds must become cheaper to compete.

RATES FALL

Existing higher-paying fixed coupons become more attractive, supporting higher prices.

PRICE ADJUSTS

The bond’s promised coupon is fixed, so market price does much of the adjustment.

PORTFOLIO EFFECT

Longer-duration holdings usually react more strongly to rate changes.

Duration: rate rise

Modified duration 5.2; yield rises 0.75 percentage points.

FORMULA

$$\text{Approx. }\%\Delta\text{Price} \approx -\text{Modified duration} \times \Delta\text{Yield}$$

WHAT ARE WE MEASURING?

Approximate sensitivity of price to a small yield change.

WORKINGS

$0.75\% = 0.0075$
$\%\Delta\text{Price} \approx -5.2 \times 0.0075$
$= -0.039 = -3.9\%$

ANSWER

≈ 3.9% price decline

PORTFOLIO MEANING

Higher duration means greater interest-rate sensitivity; the estimate ignores convexity.

Duration: rate fall

Modified duration 7.0; yield falls 0.40 percentage points.

FORMULA

$$\text{Approx. }\%\Delta\text{Price} \approx -\text{Duration} \times \Delta\text{Yield}$$

WHAT ARE WE MEASURING?

The negative sign captures the inverse price/yield relationship.

WORKINGS

$\Delta\text{Yield} = -0.40\% = -0.0040$
$-7.0 \times (-0.0040) = +0.028$
$= +2.8\%$

ANSWER

≈ 2.8% price increase

PORTFOLIO MEANING

Duration helps balance income objectives against rate sensitivity.

Credit spread

Corporate yield 6.4%; comparable government yield 4.1%.

FORMULA

$$\text{Credit spread} = \text{Corporate yield} - \text{Government yield}$$

WHAT ARE WE MEASURING?

Extra yield offered above the government benchmark.

WORKINGS

6.4% − 4.1% = 2.3 percentage points
2.3 percentage points = 230 basis points

ANSWER

2.3% = 230 bps

PORTFOLIO MEANING

A wider spread can be compensation for credit risk—or a warning of stress.

What bonds can contribute

INCOME

Coupons can provide planned cash flow.

STABILITY

High-quality short bonds may fluctuate less than equities.

DIVERSIFICATION

Some bonds can behave differently from stocks in stress.

LIABILITY MATCHING

Maturities can align with future cash needs.

LIQUIDITY

Some government markets are deep and tradable.

RISKS REMAIN

Inflation, duration and credit deterioration can still create losses.

PORTAL 20

ETFs, Commodities & Alternatives

Broaden the toolkit using pooled and alternative exposures without buying every underlying asset directly.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

What an ETF actually does

The label “ETF” describes the vehicle—not the risk inside it.

POOL

Combines investor money into a portfolio that trades on an exchange.

EXPOSURE

Can track broad markets, sectors, bonds, commodities, factors or active strategies.

HOLDINGS

Always ask what the fund really owns.

COST

Expense ratio and trading spread reduce investor return.

TRACKING

The fund may not perfectly match its benchmark.

CONCENTRATION

An ETF can still be concentrated in a few holdings or one sector.

ETF expense ratio

Investment \$8,000; expense ratio 0.25%.

FORMULA

$$\text{Approx. annual fee} = \text{Investment} \times \text{Expense ratio}$$

WHAT ARE WE MEASURING?

Approximate annual fund operating cost on the investment.

WORKINGS

0.25% = 0.0025
\$8,000 × 0.0025 = \$20

ANSWER

≈ \$20 per year

PORTFOLIO MEANING

Small annual differences can compound over long holding periods.

Tracking difference

Index return 9.2%; ETF return 8.9%.

FORMULA

$$\text{Tracking difference} = \text{ETF return} - \text{Benchmark return}$$

WHAT ARE WE MEASURING?

How the fund's return differed from the intended benchmark.

WORKINGS

8.9% − 9.2% = −0.3 percentage points

ANSWER

−0.30 percentage points

PORTFOLIO MEANING

Fees, trading costs, cash drag and imperfect replication can contribute.

What moves commodities?

SUPPLY

Production cuts, mine closures, crop failures.

DEMAND

Industry, construction, transport and consumption.

INVENTORIES

Low stocks can make shocks more powerful.

WEATHER

Important for agriculture and energy demand.

GEOPOLITICS

Sanctions, conflict and shipping disruption.

FUTURES STRUCTURE

The investment vehicle can behave differently from spot prices.

Commodity price return

Gold rises from $2,000 to $2,140 per ounce.

FORMULA

$$\text{Price return} = (\text{Ending price} - \text{Beginning price}) / \text{Beginning price} \times 100\%$$

WHAT ARE WE MEASURING?

Change in the commodity price before vehicle costs.

WORKINGS

Price change = $2,140 − $2,000 = $140
$140 / $2,000 = 0.07
0.07 × 100% = 7.0%

ANSWER

7.0%

PORTFOLIO MEANING

Commodity exposure may diversify some inflation/geopolitical risks but can be volatile.

Direct ownership vs pooled exposure

STOCKS

Individual shares vs a basket of companies.

BONDS

Specific issuer/maturity vs diversified bond fund.

GOLD

Physical bullion/storage vs liquid fund exposure.

PROPERTY

Direct property/management vs a REIT or REIT ETF.

COMMODITIES

Physical ownership can be impractical; funds may use futures.

KEY QUESTION

Choose the vehicle whose costs, liquidity and risks fit the portfolio.

Strategic Asset Allocation

Set long-term target weights based on objectives, horizon, liquidity needs and risk capacity.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Allocation is the portfolio’s architecture

WEIGHTS CREATE SCALE

A $4,000 holding means different things in a $5,000 versus $100,000 portfolio.

OBJECTIVES FIRST

Return goals must coexist with liquidity and loss capacity.

STRATEGIC

Long-term target structure.

TACTICAL

Temporary deviation based on a specific view.

DISCIPLINE

Targets make later rebalancing measurable.

NOT A FORECAST

Allocation is not a guess about next month’s winner.

Portfolio weight

$14,800 portfolio; stocks worth $4,440.

FORMULA

$$\text{Weight} = \text{Asset value} / \text{Portfolio value} \times 100\%$$

WHAT ARE WE MEASURING?

Share of total capital currently committed to stocks.

WORKINGS

$4,440 / $14,800 = 0.30
0.30 × 100% = 30%

ANSWER

30% stock weight

PORTFOLIO MEANING

Weights are the common language for comparing portfolio exposures.

Target dollars

$14,800 portfolio; target bond weight 25%.

FORMULA

$$\text{Target value} = \text{Portfolio value} \times \text{Target weight}$$

WHAT ARE WE MEASURING?

Converts a policy percentage into an implementable dollar target.

WORKINGS

25% = 0.25
$14,800 × 0.25 = $3,700

ANSWER

$3,700 in bonds

PORTFOLIO MEANING

Needed before comparing current holdings with desired allocation.

Expected portfolio return

50% stocks at 8%; 30% bonds at 4%; 20% cash at 2%.

FORMULA

$$E(Rp) = \Sigma\, w_i E(R_i)$$

WHAT ARE WE MEASURING?

Weighted combination of expected asset returns.

WORKINGS

Stocks: 0.50×8% = 4.00%
Bonds: 0.30×4% = 1.20%
Cash: 0.20×2% = 0.40%
Total = 5.60%

ANSWER

5.60% expected return

PORTFOLIO MEANING

Expected return is an assumption-based planning measure—not a promise.

Rebalancing amount

Bond target $3,700; current bonds $3,150.

FORMULA

$$\text{Trade} = \text{Target value} - \text{Current value}$$

WHAT ARE WE MEASURING?

Difference between desired and actual dollar exposure.

WORKINGS

$3,700 − $3,150 = +$550

ANSWER

Buy/add $550

PORTFOLIO MEANING

Rebalancing reconnects the real portfolio to the policy target.

Example strategic allocation

Illustrative allocation from the notes—not a universal recommendation.

STOCKS 30%

$4,440

BONDS 25%

$3,700

REAL ESTATE 15%

$2,220

DIVERSIFIED ETF 10%

$1,480

ALTERNATIVES 5%

$740

FOREX / HEDGE 5%

$740

CASH 10%

$1,480

TOTAL 100%

$14,800

PORTAL 22

Portfolio Diversification

Identify hidden concentrations by examining correlation, sector, geography, currency and liquidity.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Diversification is not “own many things”

NAMES ≠ RISKS

Ten technology stocks can still be one concentrated bet.

SECTOR

How much depends on one industry?

GEOGRAPHY

How much depends on one economy?

CURRENCY

Foreign assets add exchange-rate exposure.

LIQUIDITY

Can enough capital be accessed in a crisis?

CORRELATION

Do holdings tend to move together?

Sector concentration

Technology exposure $6,216; portfolio $14,800.

FORMULA

$$\text{Sector weight} = \text{Sector exposure} / \text{Portfolio value} \times 100\%$$

WHAT ARE WE MEASURING?

Percentage of portfolio tied to one broad sector.

WORKINGS

$6,216 / $14,800 = 0.42
0.42 × 100% = 42%

ANSWER

42% technology exposure

PORTFOLIO MEANING

Ask whether this concentration is intentional and adequately rewarded.

Correlation made simple

Diversification improves when risks do not all respond identically.

+1

Assets move perfectly together.

POSITIVE

They often move in the same direction.

0

No linear relationship in the sample.

NEGATIVE

They often move in opposite directions.

−1

Perfect opposite movement.

IMPORTANT

Correlation can change—especially during crises.

Two-asset portfolio volatility

60% A, 40% B; σA=15%, σB=10%; correlation=0.20.

FORMULA

$$\sigma p^2 = wA^2\sigma A^2 + wB^2\sigma B^2 + 2wAwB\sigma A\sigma B\rho$$

WHAT ARE WE MEASURING?

Combined risk depends on both individual volatility and correlation.

WORKINGS

Term 1 = $(0.60^2)(0.15^2)$ = 0.00810
Term 2 = $(0.40^2)(0.10^2)$ = 0.00160
Covariance term = 0.00144
Variance = 0.01114
σp = $\sqrt{0.01114}$ = 0.10555 = 10.56%

ANSWER

≈ 10.56% volatility

PORTFOLIO MEANING

Lower correlation can reduce total portfolio volatility.

What if correlation rises?

Same portfolio, but correlation rises from 0.20 to 0.90.

FORMULA

$$\sigma p^2 = wA^2\sigma A^2 + wB^2\sigma B^2 + 2wAwB\sigma A\sigma B\rho$$

WHAT ARE WE MEASURING?

The portfolio becomes riskier when assets move more closely together.

WORKINGS

Term 1 = 0.00810
Term 2 = 0.00160
Covariance term = 0.00648
Variance = 0.01618
$\sigma p = \sqrt{0.01618} = 12.72\%$

ANSWER

≈ 12.72% volatility

PORTFOLIO MEANING

Diversification benefits can weaken in a crisis.

Foreign asset + currency

Foreign stock +8%; foreign currency +5% vs home currency.

FORMULA

$$\text{Domestic return} = (1 + \text{Local return})(1 + \text{FX return}) - 1$$

WHAT ARE WE MEASURING?

Combines local asset performance with currency movement.

WORKINGS

(1+0.08)(1+0.05) − 1
1.08 × 1.05 − 1
1.134 − 1 = 0.134

ANSWER

13.4% domestic return

PORTFOLIO MEANING

Global diversification requires tracking both asset and FX exposure.

Liquidity is part of diversification

CASH / T-BILLS

Usually accessible quickly.

LARGE STOCKS / ETFs

Often liquid, but spreads can widen.

CORPORATE BONDS

Liquidity varies by issue.

DIRECT PROPERTY

Sale can take months.

PRIVATE BUSINESS

Buyer may be difficult to find.

ALTERNATIVES

Lockups or redemption limits may apply.

PORTAL 23

Add, Trim or Exit

Turn portfolio diagnosis into disciplined action rather than emotional trading.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Five core management decisions

ADD

Increase when evidence strengthens, valuation improves or position is below target.

HOLD

Keep deliberately when thesis and sizing remain appropriate.

TRIM

Reduce because valuation, concentration or risk has risen.

EXIT

Close when thesis breaks or risk becomes unacceptable.

REPLACE

Move capital to a superior opportunity.

RULE

Define reasons before price pressure creates emotion.

Rebalancing band

Target stock weight 30%; allowed deviation ±5 percentage points.

FORMULA

$$\text{Upper} = \text{Target} + \text{deviation}; \text{Lower} = \text{Target} - \text{deviation}$$

WHAT ARE WE MEASURING?

Tolerance range around the target.

WORKINGS

Upper = 30% + 5% = 35%
Lower = 30% − 5% = 25%

ANSWER

25% to 35% band

PORTFOLIO MEANING

Bands reduce needless trading while creating a pre-agreed action rule.

Trim to target

Portfolio $16,000; stock position $6,000; target 30%.

FORMULA

$$\text{Target value} = \text{Portfolio} \times \text{Target weight; Trim} = \text{Current} - \text{Target}$$

WHAT ARE WE MEASURING?

Amount required to return the position to its target weight.

WORKINGS

Target = $16,000 × 0.30 = $4,800
Trim = $6,000 − $4,800 = $1,200

ANSWER

Trim $1,200

PORTFOLIO MEANING

A trim can reduce concentration without declaring the thesis wrong.

Remaining valuation upside

Estimated value $72; current price $66.

FORMULA

$$\text{Upside} = (\text{Value} - \text{Price}) / \text{Price} \times 100\%$$

WHAT ARE WE MEASURING?

Remaining gap between market price and estimated value.

WORKINGS

Difference = $72 − $66 = $6
$6 / $66 = 0.090909
×100% = 9.09%

ANSWER

≈ 9.09% upside

PORTFOLIO MEANING

A good company may become less attractive when the valuation gap narrows.

Capital rotation

Current holding expected return 5%; new opportunity 9%.

FORMULA

$$\text{Relative advantage} = \text{New E(R)} - \text{Current E(R)}$$

WHAT ARE WE MEASURING?

Estimated expected-return advantage of the alternative.

WORKINGS

9% − 5% = 4 percentage points

ANSWER

4 percentage points

PORTFOLIO MEANING

Also consider risk, liquidity, taxes, costs and forecast uncertainty.

What can break a thesis?

ECONOMICS

Revenue or cash-flow driver weakens structurally.

ADVANTAGE

Competitive position deteriorates.

BALANCE SHEET

Debt/refinancing risk rises materially.

MANAGEMENT

Credibility or execution worsens.

REGULATION

Rules damage access, costs or profitability.

VALUATION

Price no longer compensates for risk.

CONCENTRATION

Position becomes too large.

OPPORTUNITY COST

A superior use of capital emerges.

PORTAL 24

Macroeconomics for Investors

Use the economic environment to understand forces acting across several portfolio holdings at once.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

The macro dashboard

GDP

Is economic activity accelerating or slowing?

INFLATION

Are prices rising faster or slower?

INTEREST RATES

How expensive is capital?

UNEMPLOYMENT

How strong is labor demand and household income?

FISCAL POLICY

Is government policy stimulating or restraining demand?

MONETARY POLICY

Is the central bank tightening or easing?

YIELD CURVE

What does the term structure imply?

CREDIT SPREADS

Is perceived borrower risk changing?

Approximate real rate

Nominal rate 5.5%; inflation 3.2%.

FORMULA

$$\text{Approx. real rate} \approx \text{Nominal rate} - \text{Inflation}$$

WHAT ARE WE MEASURING?

Simple inflation-adjusted rate approximation.

WORKINGS

5.5% − 3.2% = 2.3%

ANSWER

≈ +2.3%

PORTFOLIO MEANING

Higher real rates can tighten financial conditions and pressure long-duration assets.

Exact real rate

Nominal rate 5.5%; inflation 3.2%.

FORMULA

$$\text{Real} = (1 + \text{Nominal}) \,/\, (1 + \text{Inflation}) - 1$$

WHAT ARE WE MEASURING?

More precise inflation adjustment using the Fisher relationship.

WORKINGS

1.055 / 1.032 − 1
= 1.0222868 − 1
= 0.0222868 = 2.22868%

ANSWER

≈ 2.23%

PORTFOLIO MEANING

Use exact calculation when precision matters or rates/inflation are high.

Yield-curve spread

10-year yield 4.0%; 2-year yield 4.7%.

FORMULA

$$\text{Term spread} = \text{Long-term yield} - \text{Short-term yield}$$

WHAT ARE WE MEASURING?

Selected curve segment is inverted.

WORKINGS

4.0% − 4.7% = −0.7 percentage points
= −70 basis points

ANSWER

−0.70% = −70 bps

PORTFOLIO MEANING

Investigate policy and growth expectations; inversion is a signal, not a guaranteed recession timer.

How inflation can affect assets

CASH

Purchasing power can erode if yield lags inflation.

NOMINAL BONDS

Fixed payments lose real value; rising yields can reduce price.

STOCKS

Depends on pricing power, input costs and valuation.

REAL ESTATE

Rents may rise, but financing costs can also rise.

COMMODITIES

Some react strongly to supply shocks and inflation.

FOREX

Relative inflation and policy expectations affect currencies.

Economic cycle: ask different questions

EXPANSION

Growth strong: are cyclicals and credit risks underpriced?

LATE CYCLE

Inflation/rates may rise: is leverage becoming dangerous?

SLOWDOWN / RECESSION

Which earnings are cyclical? Is liquidity sufficient?

RECOVERY

Where are prices still reflecting recession conditions?

PORTAL 25

Earnings, News & New Information

Separate thesis-changing evidence from market noise by comparing new facts with prior expectations.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Markets react to surprise—not simply “good” or “bad”

EXPECTATIONS

A record result can disappoint if investors expected even more.

SOURCE

Prefer primary, credible evidence.

CASH FLOW

Does the news change future cash generation?

RISK

Does it change the probability or severity of loss?

VALUATION

Does the value estimate need revision?

DURATION

Is the change temporary or structural?

Earnings surprise

Actual EPS $2.10; consensus expected $1.95.

FORMULA

$$\text{Surprise \%} = (\text{Actual} - \text{Expected}) / |\text{Expected}| \times 100\%$$

WHAT ARE WE MEASURING?

Reported EPS exceeded consensus by about 7.69%.

WORKINGS

Difference = $2.10 − $1.95 = $0.15
$0.15 / $1.95 = 0.076923
×100% = 7.69%

ANSWER

≈ +7.69%

PORTFOLIO MEANING

Now investigate whether the beat is sustainable or driven by temporary/accounting factors.

Guidance revision

Old revenue guide \$980m–\$1.02bn; new \$1.04bn–\$1.10bn.

FORMULA

$$\text{Midpoint} = (\text{Low} + \text{High})/2;\ \text{Change} = (\text{New} - \text{Old})/\text{Old} \times 100\%$$

WHAT ARE WE MEASURING?

Management raised its central revenue expectation materially.

WORKINGS

Old midpoint = (\$980m+\$1,020m)/2 = \$1,000m
New midpoint = (\$1,040m+\$1,100m)/2 = \$1,070m
Change = \$70m/\$1,000m ×100% = 7.0%

ANSWER

Guidance midpoint +7.0%

PORTFOLIO MEANING

A weak-growth thesis may need revision if the new guidance is credible.

Event return

Stock \$50 before earnings; \$46 after.

FORMULA

$$\text{Event return} = (\text{Post-event price} - \text{Pre-event price}) / \text{Pre-event price} \times 100\%$$

WHAT ARE WE MEASURING?

Measures the market price reaction around the event.

WORKINGS

Change = \$46 − \$50 = −\$4
−\$4 / \$50 = −0.08
×100% = −8.0%

ANSWER

−8.0%

PORTFOLIO MEANING

It does not prove the market is right. Identify what expectation changed.

Information triage: seven questions

1

What exactly changed?

2

Is the source primary and credible?

3

Was it already expected?

4

Does it alter cash flow, risk, valuation or financing?

5

Temporary or structural?

6

One holding or the whole portfolio?

7

Does the action change: ADD, HOLD, TRIM, EXIT—or no action?

DISCIPLINE

A price move alone is not a thesis.

PORTAL 26

Risk Management & Stress Testing

Measure different dimensions of risk and test how the portfolio might behave under adverse scenarios.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Risk has more than one meaning

VOLATILITY

How widely returns vary.

BETA

Sensitivity to broad market movement.

DRAWDOWN

Peak-to-trough loss.

SHARPE

Excess return per unit of volatility.

VaR

Model-based loss threshold under assumptions.

LEVERAGE

Exposure controlled per dollar of equity.

LIQUIDITY

Ability to transact at a reasonable price.

STRESS TEST

What happens under a chosen adverse scenario?

Average return

Monthly returns: 2%, −1%, 3%, 0%, 1%.

FORMULA

$$\text{Arithmetic mean} = \Sigma R_i \;/\; n$$

WHAT ARE WE MEASURING?

Central arithmetic return across observations.

WORKINGS

Sum = 2% −1% +3% +0% +1% = 5%
n = 5
Mean = 5% / 5 = 1%

ANSWER

1% average monthly return

PORTFOLIO MEANING

Average return alone does not tell you how unstable the path was.

Sample volatility

Returns: 2%, −1%, 3%, 0%, 1%; mean = 1%.

FORMULA

$$s^2 = \Sigma(R_i - \bar{R})^2/(n-1);\ s = \sqrt{s^2}$$

WHAT ARE WE MEASURING?

Historical spread of returns around the mean.

WORKINGS

Deviations: 1%, −2%, 2%, −1%, 0%
Decimals squared: .0001,.0004,.0004,.0001,0
Sum = 0.0010; variance = 0.0010/4 = 0.00025
s = √0.00025 = 0.015811 = 1.58%

ANSWER

≈ 1.58% monthly volatility

PORTFOLIO MEANING

Higher volatility means wider variation, but does not capture every form of risk.

Beta interpretation

Stock beta 1.30; market rises 4%.

FORMULA

$$\text{Approx. market-driven move} \approx \text{Beta} \times \text{Market move}$$

WHAT ARE WE MEASURING?

Simplified market-related sensitivity estimate.

WORKINGS

$1.30 \times 4\% = 5.2\%$

ANSWER

$\approx +5.2\%$

PORTFOLIO MEANING

Beta is not an exact return forecast and excludes company-specific news.

Drawdown

Portfolio peak $18,000; trough $14,400.

FORMULA

$$\text{Drawdown} = (\text{Trough} - \text{Peak}) / \text{Peak} \times 100\%$$

WHAT ARE WE MEASURING?

Loss from the prior peak to the trough.

WORKINGS

Loss = $14,400 − $18,000 = −$3,600
−$3,600/$18,000 = −0.20
×100% = −20%

ANSWER

−20% drawdown

PORTFOLIO MEANING

Drawdown expresses the decline an investor actually had to endure.

Recovery after drawdown

Recover from $14,400 back to $18,000.

FORMULA

$$\text{Recovery return} = \text{Peak} / \text{Trough} - 1$$

WHAT ARE WE MEASURING?

Losses and recoveries are asymmetric because recovery starts from a smaller base.

WORKINGS

$18,000/$14,400 = 1.25
1.25 − 1 = 0.25
= 25%

ANSWER

25% required gain

PORTFOLIO MEANING

A 20% loss requires a 25% gain to get back to the original peak.

Sharpe ratio

Portfolio return 9%; risk-free 3%; volatility 12%.

FORMULA

$$\text{Sharpe} = (R_p - R_f) / \sigma_p$$

WHAT ARE WE MEASURING?

Excess return earned per unit of volatility.

WORKINGS

Excess return = 9% − 3% = 6%
0.06 / 0.12 = 0.50

ANSWER

0.50

PORTFOLIO MEANING

Compare only when periods, inputs and methodology are consistent.

Simple 95% one-day VaR

Portfolio \$100,000; one-day volatility 1.2%; z=1.645; zero mean.

FORMULA

$$\text{VaR} \approx \text{Portfolio value} \times z \times \sigma$$

WHAT ARE WE MEASURING?

Model-based 95% one-day loss threshold under normal assumptions.

WORKINGS

$\sigma = 0.012$
$1.645 \times 0.012 = 0.01974$
$\$100{,}000 \times 0.01974 = \$1{,}974$

ANSWER

≈ \$1,974

PORTFOLIO MEANING

There is still a 5% tail beyond the threshold; VaR is not worst-case loss.

Leverage

Investor equity \$20,000; gross exposure \$30,000.

FORMULA

$$\text{Leverage} = \text{Gross exposure} / \text{Equity}$$

WHAT ARE WE MEASURING?

Investor controls \$1.50 of gross exposure for each \$1 of equity.

WORKINGS

\$30,000/\$20,000 = 1.5

ANSWER

1.5× leverage

PORTFOLIO MEANING

Leverage magnifies gains, losses and potential forced-selling risk.

Recession stress test

$6,000 stocks −20%; $4,000 bonds +5%; $3,000 property −15%; $1,800 cash 0%.

FORMULA

$$\text{Scenario P/L} = \Sigma(\text{Position value} \times \text{Scenario return})$$

WHAT ARE WE MEASURING?

Estimated portfolio dollar loss under the chosen shocks.

WORKINGS

Stocks = −$1,200
Bonds = +$200
Property = −$450
Cash = $0
Total = −$1,450

ANSWER

−$1,450

PORTFOLIO MEANING

Stress tests reveal which exposures drive loss and which provide protection.

Scenario return

Starting portfolio \$14,800; scenario P/L −\$1,450.

FORMULA

$$\text{Scenario return} = \text{Scenario P/L} \,/\, \text{Starting value} \times 100\%$$

WHAT ARE WE MEASURING?

Portfolio-level percentage loss under the scenario.

WORKINGS

$-\$1{,}450/\$14{,}800 = -0.097973$
$\times 100\% = -9.80\%$

ANSWER

$\approx -9.80\%$

PORTFOLIO MEANING

Compare the loss with risk tolerance, liquidity needs and mandate.

Stress more than prices

RECESSION

Revenue, credit spreads, property and unemployment.

INFLATION SHOCK

Bond prices, margins and pricing power.

RATE SHOCK

Duration, mortgages and valuations.

CURRENCY SHOCK

Foreign holdings and hedges.

COMPANY SHOCK

What if the largest holding falls 40%?

LIQUIDITY SHOCK

What if private/property assets cannot be sold for months?

PORTAL 27

Green City Crisis Challenge

Combine market, rate, credit, property, currency and company shocks—and make decisions under pressure.

UNDERSTAND → CALCULATE → INTERPRET → MANAGE

Crisis triage: use this order

1 LIQUIDITY

Are there near-term cash obligations?

2 FORCED RISKS

Margin calls, refinancing, covenants or redemption pressure?

3 THESIS BREAKS

Which assumptions are no longer valid?

4 CONCENTRATION

Which positions now dominate risk?

5 STRESS LOSS

What does the scenario do to capital?

6 VALUATION

Have prices fallen more than fundamentals?

7 ACTION

Add, hold, trim, exit, replace or hedge.

8 RECORD

Write the reason before acting.

Integrated crisis scenario

The objective is not to avoid every loss. It is to respond rationally to several shocks at once.

RATES

Policy rates unexpectedly +1.0 percentage point.

EQUITIES

Broad equity market −15%.

PROPERTY

Portfolio property value −10%.

CREDIT

Corporate spreads widen.

CURRENCY

Home currency appreciates 6%.

COMPANY

A major holding cuts guidance.

Crisis duration loss

$3,700 bond allocation; modified duration 5.0; yield +1.0%.

FORMULA

$$\%\Delta\text{Price} \approx -\text{Duration} \times \Delta\text{Yield}; \text{Dollar P/L} \approx \text{Position} \times \%\Delta\text{Price}$$

WHAT ARE WE MEASURING?

Approximate interest-rate loss on the bond allocation.

WORKINGS

$\Delta\text{Yield} = 0.010$
$\%\Delta\text{Price} \approx -5.0 \times 0.010 = -5.0\%$
$\$3{,}700 \times (-0.05) = -\$185$

ANSWER

≈ −$185

PORTFOLIO MEANING

Judge the loss alongside the bond’s liquidity/stabilizing role and higher future reinvestment yields.

Currency shock

Foreign asset local return 0%; foreign currency −6% vs home currency.

FORMULA

$$\text{Domestic return} = (1+\text{Local})(1+\text{FX}) - 1$$

WHAT ARE WE MEASURING?

Currency translation creates a loss even though local asset price is unchanged.

WORKINGS

$(1+0)(1-0.06) - 1$
$= 0.94 - 1 = -0.06$

ANSWER

−6.0%

PORTFOLIO MEANING

FX risk can matter even when asset fundamentals have not changed.

Weights change during crisis

Defensive holding stays $2,500; portfolio falls to $12,500.

FORMULA

$$\text{New weight} = \text{New position value} / \text{New portfolio value} \times 100\%$$

WHAT ARE WE MEASURING?

A stable holding becomes a larger share when the rest of the portfolio falls.

WORKINGS

$2,500/$12,500 = 0.20
×100% = 20%

ANSWER

20% new weight

PORTFOLIO MEANING

Rebalance using updated values, not old percentages.

Crisis decision grid

LONG-DURATION BOND

Rate shock → quantify duration loss before acting.

SHORT HIGH-QUALITY BOND

May preserve liquidity and stability.

GROWTH STOCK

Separate rate-driven repricing from thesis damage.

WEAK COMPANY

Guidance cut may represent thesis deterioration.

DIRECT PROPERTY

Check debt service and refinancing capacity.

FOREIGN ASSET

Separate asset return from FX translation.

CASH

Provides optionality when other assets are stressed.

RULE

Falling price alone is not a sell signal.

Green City integrated decision workflow

Use one repeatable process across every holding and every portal.

1 OBSERVE

What changed in price, economics or information?

2 CALCULATE

Use the relevant formula and show workings.

3 INTERPRET

Translate the number into portfolio meaning.

4 COMPARE

Targets, benchmarks, alternatives and scenarios.

5 ACT

Add, hold, trim, exit, replace, hedge or rebalance.

6 REVIEW

Define what evidence would change the decision next time.

Calculation self-check

A correct number is only the middle of the analysis.

FORMULA

Did I choose the correct relationship?

UNITS

Are dollars, percentages, years and rates consistent?

PERCENTAGES

Did I convert to decimals when multiplying?

WORKINGS

Can another learner follow each step?

INTERPRETATION

Can I explain the answer in one sentence?

BENCHMARK

What should this number be compared with?

LIMITATION

What does the calculation not tell me?

DECISION

How does the result affect portfolio action?

Green City readiness checklist

Strong capability means you can explain, calculate, interpret and act.

BONDS

Coupon, yield, HPR, duration and credit risk.

ETFs / ALTERNATIVES

Costs, holdings, tracking and exposure.

ALLOCATION

Weights, target dollars and expected portfolio return.

DIVERSIFICATION

Concentration, correlation, FX and liquidity.

ACTIVE MANAGEMENT

Add, hold, trim, exit and replace.

MACRO

Rates, inflation, yield curve and cycles.

NEW INFORMATION

Expectations, earnings, guidance and event reaction.

RISK

Volatility, beta, drawdown, Sharpe, VaR and stress tests.

GREEN CITY COMPLETION

Manage the portfolio—not the headlines.

A strong Green City learner can adapt to changing markets without abandoning a disciplined process.

1 Measure exposures	**2** Show calculations	**3** Interpret risk
4 Use new information	**5** Rebalance deliberately	**6** Act under crisis pressure

NEXT: YELLOW CITY — professional portfolio construction and optimization.